

Jeffrey Guo

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EDUCATION

University of Southern California

Masters of Science in Computer Science

Los Angeles, California

January 2024 – December 2025

- GPA: 3.57/4.0
- Relevant Coursework: Natural Language Processing, Deep Learning, Multimodal Probabilistic Learning of Human Communication, Data Structures and Algorithms, Database Systems

University of British Columbia

Bachelor of Science in Computer Science

Vancouver, British Columbia

September 2018 – May 2023

- GPA: 3.7/4.0
- Relevant Coursework: Machine Learning and Data Mining, Intermediate Algorithm Design and Analysis, Introduction to Artificial Intelligence, Computer Vision

EXPERIENCE

Programming Instructor

Sager Education

Sep. 2021 – Apr. 2022, Oct. 2023 – July 2024

Vancouver, British Columbia

- Designed and taught K-12 curriculum in Python, Java, and full-stack web development, authoring interactive projects and presentations covering OOP, Data Structures, and software design principles.

PROJECTS

AI Video Content Repurposing Engine | *Python, React, Google Cloud, Docker* November 2025 – January 2026

- Architected a serverless full-stack platform transforming YouTube videos into structured, SEO-optimized blog posts using Gemini 2.0 Flash and Firebase, supporting videos up to 60 minutes in length.
- Engineered a low-latency backend pipeline on Cloud Run with stream-based audio extraction, processing long-form content within a strict 512MB RAM constraint.
- Implemented an OAuth2 authentication flow to securely authorize server-side media requests, successfully bypassing anti-bot IP blocking mechanisms.

Audio Visual Look-At-Me Model | *Python, PyTorch, Scikit*

January 2025 – April 2025

- Developed a multi-modal gaze detection model identifying camera-directed attention in egocentric video, achieving ~85% accuracy on held-out test data.
- Processed and aligned audio-visual data from 150 videos, constructing a cropped-head dataset for model training
- Fused ResNet-encoded visual features with Whisper-encoded audio embeddings to train a Conversational Attention classifier, outperforming a vision-only baseline.

Video Keyframes to Summary Neural Network | *Python, PyTorch*

September 2024 – December 2024

- Designed an end-to-end pipeline to automatically generate video summaries by extracting keyframes, captioning them, and synthesizing a final summary via a fine-tuned T5 model.
- Benchmarked 3 keyframe extraction methods (Katna, Video-kf, optical flow) to identify the highest-quality frame selector for downstream captioning.
- Outperformed baseline and BART across all metrics — ROUGE-1: 0.265 vs. 0.208 baseline, BERTScore-F1: 0.0801 vs. 0.0124 baseline — validated using an LLM-as-a-judge evaluator built with Google Gemini.

Multi-Agent Cooperative Training | *C#, Python, Unity*

September 2024 – December 2024

- Leveraged Unity ML-Agents MA-POCA algorithm, designing a custom environment and reward structure to maximize emergent cooperative behavior.
- Trained agents over 5,000+ episodes via Multi-Agent Reinforcement Learning, analyzing behavioral evolution across runs to evaluate reward shaping strategies.

TECHNICAL SKILLS

Languages: Python, JavaScript, Typescript, Java, C#, C/C++, HTML/CSS, SQL

Frameworks/Libraries: PyTorch, Scikit, React, FastAPI, Vite, Node.js, Next.js, Express, pandas, NumPy, Matplotlib, react-bootstrap, MongoDB

Technologies: Git, PowerBI, AWS, Docker, PostgreSQL, MongoDB, Google Cloud Platform, Firebase, OAuth2